

Business Requirement Document

Shopping Cart Module

ShopSphere E-Commerce Platform — ShopSphere Platform Modernization Program (FY2026)

Document Title	Business Requirement Document — Shopping Cart Module
Document ID	BRD-EC-SCT-003
Version	1.0
Author	Vikram Singh, Senior Business Analyst
Date	28-Jan-2026
Status	Under Review

1. Document Information

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2. Executive Summary

The Shopping Cart Module is a core transactional component of the ShopSphere E-Commerce Platform, serving approximately 2 million monthly active users across India, UAE, and Southeast Asia. As the primary staging area between product discovery and checkout, the cart directly influences conversion rate, average order value, and Gross Merchandise Value (GMV). This document defines the business, functional, and non-functional requirements for re-architecting the Shopping Cart Module as part of the ShopSphere Platform Modernization Program for fiscal year 2026. The modernization effort is driven by rising cart abandonment rates (currently benchmarked at 68% across web and mobile channels), inconsistent cart state across devices, and stale pricing/stock data that erodes customer trust at the point of checkout handoff. The redesigned module will deliver real-time price and inventory recalculation, seamless cross-device cart persistence, intelligent coupon stacking, and automated abandoned-cart re-engagement, positioning ShopSphere to meaningfully reduce abandonment and improve checkout conversion during both steady-state trading and peak sale events such as Diwali and regional Black Friday campaigns.

3. Business Objective

The primary business objective is to reduce shopping cart abandonment and improve the reliability of the pre-checkout experience so that more browsing sessions convert into completed orders. Specifically, the Shopping Cart Module modernization aims to: (a) eliminate revenue leakage caused by stale price or stock information discovered late in the funnel; (b) enable customers to seamlessly resume an in-progress cart regardless of device or channel, increasing session-to-purchase continuity; (c) provide a robust promotions and coupon-stacking framework that increases average order value without eroding margin through uncontrolled discount combinations; and (d) automate abandoned-cart detection and re-engagement to recover otherwise-lost GMV. These objectives directly support the broader ShopSphere Platform Modernization Program goal of increasing platform-wide conversion rate by 15% and reducing customer-reported pricing/stock discrepancies by 90% within FY2026.

4. Project Scope

In Scope

- Add, update, and remove cart items (SKU-level quantity changes, size/variant changes).

- Save-for-later and wishlist crossover functionality, allowing items to move between cart, save-for-later, and wishlist states.
- Cart persistence across sessions and devices, synchronizing state between web and mobile app for authenticated users.
- Promo/coupon code application, including validation and coupon-stacking rules.
- Real-time price and stock recalculation against the Pricing & Promotions engine and Inventory Management System (IMS).
- Cart abandonment detection logic and triggering of reminder notifications (email/SMS/push).
- Multi-currency price display for UAE (AED) and Southeast Asia (SGD, MYR, PHP, THB, VND) storefronts.

Out of Scope

- Checkout workflow and payment processing (addressed in BRD-EC-CKO-002 — Checkout & Payment Module).
- Build of the loyalty points redemption engine (loyalty balance display only; redemption logic is a separate initiative).
- Gift-wrapping fulfillment logistics and warehouse-side packaging workflows.

5. Stakeholders

Name	Role	Department	Responsibility
Rajesh Mehta	VP of E-Commerce	Business	Executive Sponsor
Ananya Sharma	Product Owner, ShopSphere Platform	Product	Prioritization & sign-off
Vikram Singh	Senior Business Analyst	Business Analysis	Requirements owner
Sarah Chen	Enterprise IT Architect	IT/Architecture	Technical feasibility
Michael Torres	QA Lead	Quality Assurance	Test strategy & sign-off
Priya Nair	UX Lead	Design	User experience
David Kim	DevOps Lead	Engineering/Infra	Deployment & monitoring
Fatima Al-Sayed	Compliance & Security Officer	Risk & Compliance	Regulatory compliance

6. Business Requirements

ID	Business Requirement
BR-001	The platform must reduce overall cart abandonment rate by providing an accurate, trustworthy, and low-friction cart experience across all channels.
BR-002	The platform must allow customers to access an identical, up-to-date cart regardless of whether they last interacted with it on the ShopSphere website or the ShopSphere mobile app.
BR-003	The platform must prevent customers from proceeding to checkout with items that are no longer available or whose price has changed, by surfacing conflicts within the cart itself.
BR-004	The platform must support a flexible and controllable coupon/promotion stacking model that increases average order value while protecting margin.
BR-005	The platform must proactively re-engage customers who abandon a cart, in order to recover potentially lost GMV.
BR-006	The platform must display cart pricing in the customer's local currency for UAE and Southeast Asia markets to build purchase confidence.
BR-007	The platform must give customers an easy mechanism to defer purchase intent via save-for-later and wishlist without losing selected items.
BR-008	The platform must provide business stakeholders with visibility into cart-level behavior (abandonment, coupon usage, conflict frequency) to inform merchandising and marketing decisions.

7. Functional Requirements

7.1 Cart Item Management

- **FR-001** The system shall allow an authenticated or guest customer to add a SKU/variant combination to the cart from the PDP or PLP, with the selected quantity and variant attributes (size, color, etc.) recorded against the cart line item.

- **FR-002** The system shall allow a customer to increase or decrease the quantity of any cart line item, subject to available stock and any configured maximum-order-quantity rule for that SKU.
- **FR-003** The system shall allow a customer to remove one or more line items from the cart individually or clear the cart entirely, with a confirmation step for full cart clearance.

7.2 Save-for-Later and Wishlist Crossover

- **FR-004** The system shall allow a customer to move a cart line item to a "Saved for Later" list without deleting it, preserving variant selection and quantity.
- **FR-005** The system shall allow a customer to move an item between the cart and their wishlist in either direction, and shall reflect wishlist-to-cart moves as a new cart addition subject to current price and stock.

7.3 Cart Persistence and Cross-Device Sync

- **FR-006** The system shall persist an authenticated customer's cart server-side and synchronize it within 5 seconds across web and mobile app sessions upon login.
- **FR-007** The system shall merge a guest cart into the customer's persisted cart at the point of login/registration, applying a defined conflict-resolution rule (highest quantity wins per SKU) when the same SKU exists in both carts.
- **FR-008** The system shall retain a guest (unauthenticated) cart for a minimum of 30 days using a device/browser-scoped identifier.

7.4 Pricing, Promotions, and Multi-Currency

- **FR-009** The system shall recalculate the price, applicable discounts, and taxes of every cart line item in real time whenever the cart is viewed or modified, by querying the Pricing & Promotions engine.
- **FR-010** The system shall allow a customer to enter a promo/coupon code at the cart level, validate it against eligibility rules (minimum cart value, SKU/category restriction, customer segment, usage limit), and display the resulting discount before checkout handoff.
- **FR-011** The system shall support configurable coupon-stacking rules that determine whether multiple coupons/promotions may be applied simultaneously to the same cart, and shall reject or flag combinations that violate the configured stacking policy.
- **FR-012** The system shall display all cart prices, subtotals, and totals in the local currency associated with the customer's storefront (INR, AED, SGD, MYR, PHP, THB, VND) using the current exchange rate published by the Pricing & Promotions engine.

7.5 Stock Validation and Conflict Handling

- **FR-013** The system shall validate real-time stock availability for every cart line item against the Inventory Management System (IMS) prior to allowing checkout handoff.
- **FR-014** The system shall clearly flag any cart line item that has gone out of stock, reduced below the requested quantity, or been discontinued since it was added, and shall block checkout handoff for the affected line item until the customer adjusts quantity or removes the item.
- **FR-015** The system shall present the customer with an alternative-SKU recommendation (e.g., a similar in-stock product) when an item in the cart becomes unavailable.

7.6 Cart Abandonment Detection and Reminders

- **FR-016** The system shall classify a cart as "abandoned" when it has one or more items and no cart or checkout activity has occurred for a configurable idle period (default 60 minutes).
- **FR-017** The system shall trigger an abandoned-cart notification sequence (email, SMS, and/or push notification) to an identifiable customer through the Notification Service, with the sequence and channel configurable by the marketing team.
- **FR-018** The system shall record abandonment and recovery events (reminder sent, cart resumed, order completed) and make them available to the Analytics/BI platform.

8. Non-Functional Requirements

Performance

- **NFR-001** Cart view and update API calls shall respond in under 2 seconds at the 95th percentile under normal load.

- **NFR-002** Real-time price/stock recalculation for a cart with up to 25 line items shall complete in under 1.5 seconds at the 95th percentile.
- **NFR-003** Cross-device cart synchronization shall complete within 5 seconds of a change being made on either channel.

Security

- **NFR-004** The Shopping Cart Module shall comply with OWASP Top 10 secure coding practices for all customer-facing APIs.
- **NFR-005** Cart data containing PII (customer ID, delivery pincode used for stock check) shall be encrypted using AES-256 at rest and TLS 1.2 or higher in transit.
- **NFR-006** Coupon codes and pricing logic shall be validated server-side only; client-supplied price or discount values shall never be trusted or persisted.

Availability

- **NFR-007** The Shopping Cart Module shall maintain 99.9% uptime for customer-facing cart services.
- **NFR-008** Planned maintenance windows shall not overlap with peak sale event periods (e.g., Diwali, regional Black Friday).

Scalability

- **NFR-009** The cart service shall scale to support up to 10x normal traffic volume during peak sale events without degradation beyond stated performance targets.
- **NFR-010** The cart data store shall support horizontal scaling to accommodate growth in monthly active users across all three operating regions.

Reliability

- **NFR-011** The cart service shall meet a Recovery Point Objective (RPO) of 15 minutes and a Recovery Time Objective (RTO) of 1 hour.
- **NFR-012** No cart data loss shall occur for authenticated customers in the event of a single availability-zone failure.

9. User Stories

- **US-001** As a returning customer, I want my cart to look the same on my phone as it did on my laptop, so that I don't have to re-add items I already selected.
- **US-002** As a shopper, I want to see immediately if an item's price changed since I added it to my cart, so that I am not surprised at checkout.
- **US-003** As a shopper, I want to be notified if an item in my cart goes out of stock, so that I can choose an alternative before I try to check out.
- **US-004** As a shopper, I want to apply a coupon code to my cart and see the discount instantly, so that I know exactly what I'll pay.
- **US-005** As a shopper who is not ready to buy, I want to move an item to "Saved for Later," so that it doesn't clutter my active cart but isn't lost either.
- **US-006** As a shopper in Dubai, I want to see cart prices in AED, so that I can understand the actual cost without converting currency manually.
- **US-007** As a customer who left items in my cart, I want to receive a reminder email or push notification, so that I remember to complete my purchase.
- **US-008** As a marketing manager, I want to configure which coupons can be combined, so that we can run promotions without unintended margin loss.

10. Acceptance Criteria

AC ID	Related ID	Criteria	Expected Result
AC-001	FR-001	Given a customer on a PDP, when they select a variant and click "Add to Cart," then the item appears in the cart with correct variant and quantity.	Cart line item created accurately.
AC-002	FR-006	Given a customer adds an item on mobile, when they log in on web within 5 seconds, then the same item appears in the web cart.	Cart synced across devices within SLA.

AC-003	FR-009	Given a price change occurs in the Pricing engine, when the customer reopens their cart, then the updated price is displayed with a visible change indicator.	Cart reflects current price, not stale price.
AC-004	FR-014	Given a cart item's stock drops to zero, when the customer views the cart, then the item is flagged as out-of-stock and checkout is blocked for that line.	Customer cannot proceed with unavailable item.
AC-005	FR-010	Given a valid, eligible coupon code, when the customer applies it, then the discount is calculated and displayed within 2 seconds.	Discount reflected accurately in cart total.
AC-006	FR-011	Given two coupons that are not configured to stack, when a customer attempts to apply both, then the system rejects the second coupon with an explanatory message.	Invalid stacking combination prevented.
AC-007	FR-016/FR-017	Given a cart is idle for 60 minutes with items present, when the idle threshold is reached, then an abandoned-cart reminder is triggered via the Notification Service.	Reminder sent through configured channel.
AC-008	FR-012	Given a customer browsing the UAE storefront, when they view their cart, then all prices are displayed in AED using the current exchange rate.	Correct localized currency displayed.
AC-009	FR-004	Given an item in the active cart, when the customer selects "Save for Later," then the item moves to the saved list and is removed from the active cart total.	Cart total recalculated excluding saved item.
AC-010	FR-007	Given a guest cart and an existing account cart with an overlapping SKU, when the guest logs in, then the merged cart retains the higher quantity for that SKU.	Cart merge conflict resolved per rule.

11. Business Rules

- **RULE-001** A cart line item quantity shall never exceed the lesser of available stock (ATP) or the configured maximum-order-quantity for that SKU.
- **RULE-002** Only one "primary" percentage-off or flat-off coupon may be applied per cart unless the coupon is explicitly flagged as stackable in the Pricing & Promotions engine.
- **RULE-003** A guest cart shall be automatically purged after 30 days of inactivity.
- **RULE-004** When merging a guest cart into an authenticated customer's cart, the line item with the higher quantity for a given SKU shall be retained.
- **RULE-005** A cart shall be classified as abandoned only if it contains at least one item and has had no activity for 60 minutes; carts with zero items are excluded from abandonment tracking.
- **RULE-006** Price and stock shall be re-validated at the moment of checkout handoff, regardless of when they were last checked during the cart session.
- **RULE-007** Multi-currency prices displayed in the cart shall use the exchange rate in effect at the time of cart view, not the rate at the time the item was added.
- **RULE-008** An item flagged out-of-stock shall remain visible in the cart (not silently removed) until the customer explicitly removes it or replaces it with an alternative.

12. Process Flow Description

Primary Flow — Add to Cart through Checkout Handoff:

1. Customer browses a PLP or PDP and selects a product variant and quantity.
2. Customer clicks "Add to Cart"; the system validates stock availability via the IMS and adds the line item to the cart.
3. The system queries the Pricing & Promotions engine to calculate the current price, applicable taxes, and any auto-applied promotions for the line item.
4. The cart is persisted server-side (for authenticated customers) or in a device-scoped guest cart, and the updated cart icon/count is displayed to the customer.
5. Customer optionally continues shopping, adding further items, moving items to "Saved for Later," or applying a promo/coupon code.
6. Each time the cart is viewed, the system re-validates price and stock for every line item in real time.
7. Customer proceeds to checkout; the system performs a final stock and price re-validation across all cart line items before handing off to the Checkout & Payment Module.
8. If all line items are valid and in stock, the cart is locked for checkout and control passes to the Checkout & Payment Module.

Exception Flow — Stock Conflict Detected at Checkout Handoff:

1. During final re-validation (Step 7 above), the system detects that one or more line items are out of stock or have insufficient quantity available.
2. The system halts checkout handoff and returns the customer to the cart view, with the affected line item(s) clearly flagged.
3. The system presents the customer with options: reduce quantity to the available amount, select an alternative recommended SKU, or remove the item.
4. Customer resolves the conflict for each flagged item.
5. The system re-runs stock and price validation; if all conflicts are resolved, checkout handoff proceeds as in the primary flow.

13. Assumptions

- The Inventory Management System (IMS) exposes a real-time (sub-second) stock query API suitable for synchronous cart validation.
- The Pricing & Promotions engine is the single source of truth for price, tax, and promotion eligibility, and is available to the cart service via API.
- Customers who are logged in on multiple devices simultaneously will accept "last write wins" behavior for concurrent cart edits.
- Currency exchange rates are refreshed at least once daily and published by the Pricing & Promotions engine.
- The Notification Service (SendGrid Email API, Twilio SMS Gateway, Firebase Cloud Messaging) is available and pre-integrated with customer contact preferences.
- Marketing has the operational capability to configure abandoned-cart reminder cadence without engineering involvement.

14. Constraints

- The module must be delivered within the FY2026 ShopSphere Platform Modernization Program budget and timeline.
- Cart services must be built to withstand 10x traffic surges during announced peak sale events without dedicated infrastructure re-provisioning per event.
- All customer PII within cart data must comply with regional data protection regulations across India, UAE, and Southeast Asia jurisdictions.
- Legacy cart data migration must complete without customer-visible downtime exceeding the defined maintenance window.

15. Dependencies

- Inventory Management System (IMS) and NetSuite WMS for real-time stock and Available-to-Promise (ATP) data.
- Pricing & Promotions engine for price calculation, tax computation, and coupon eligibility rules.
- Notification Service (SendGrid Email API, Twilio SMS Gateway, Firebase Cloud Messaging) for abandoned-cart reminders.
- Checkout & Payment Module (BRD-EC-CKO-002) as the downstream consumer of the validated cart at handoff.
- Salesforce CRM for feeding cart activity into the Customer 360 profile for marketing segmentation.

16. Risks

Risk ID	Description	Likelihood	Impact	Mitigation
RISK-001	Stale stock data causes customers to reach checkout with unavailable items, damaging trust.	Medium	High	Real-time IMS validation at add-to-cart, cart view, and checkout handoff (FR-013, RULE-006).
RISK-002	Uncontrolled coupon stacking erodes margin during peak sale events.	Medium	High	Configurable stacking rules enforced server-side (FR-011, RULE-002).
RISK-003	Cart service cannot scale during Diwali/Black Friday traffic spikes, causing outages.	Medium	High	Load testing to 10x traffic, auto-scaling infrastructure, DevOps monitoring (NFR-009).
RISK-004	Cross-device sync conflicts lead to lost items or duplicated line items.	Medium	Medium	Defined merge conflict rule (RULE-004) and automated regression testing.

RISK-005	Excessive abandoned-cart notifications lead to customer opt-out or spam complaints.	Low	Medium	Configurable cadence caps and customer notification preference honoring.
RISK-006	Currency exchange rate staleness leads to incorrect cart totals for UAE/Southeast Asia customers.	Low	Medium	Daily minimum rate refresh cycle enforced via Pricing & Promotions engine (Assumption).

17. Data Requirements

The Shopping Cart Module captures and stores cart-level and line-item-level data required to support persistence, pricing accuracy, and abandonment tracking. Cart data linked to an identifiable customer is treated as PII and is subject to encryption at rest (AES-256) and in transit (TLS 1.2+), consistent with the platform-wide security baseline. Guest cart data is retained for 30 days (RULE-003); authenticated customer cart history (completed, abandoned, or merged) is retained for 13 months to support analytics and marketing re-engagement, after which it is purged or anonymized in line with data retention policy.

Data Field	Description	PII
Cart ID	Unique identifier for the cart instance (guest or authenticated).	No
Customer ID	Reference to the authenticated customer profile, where applicable.	Yes
SKU / Variant ID	Identifies the specific product and variant added to the cart.	No
Quantity	Number of units of the SKU/variant in the cart line item.	No
Unit Price / Currency	Price at time of last recalculation, in the applicable local currency.	No
Applied Coupon Code(s)	Coupon(s) applied to the cart, with computed discount amount.	No
Delivery Pincode/Area	Used for real-time stock/serviceability check; linked to customer location.	Yes
Cart Status	Active, Saved-for-Later, Abandoned, Merged, or Converted.	No
Last Activity Timestamp	Used to compute abandonment idle threshold.	No

18. Integration Requirements

System	Purpose	Direction	Data Exchanged
Inventory Management System (IMS) / NetSuite WMS	Real-time stock and ATP validation before checkout handoff.	Bi-directional	SKU stock levels, reservation confirmations.
Pricing & Promotions Engine	Price, tax, discount, and coupon eligibility calculation.	Bi-directional	SKU price, applicable promotions, coupon validation results.
Notification Service (SendGrid, Twilio, Firebase Cloud Messaging)	Abandoned-cart email/SMS/push reminder delivery.	Outbound	Customer contact reference, cart summary, reminder trigger event.
Salesforce CRM	Feed cart activity into Customer 360 profile for marketing segmentation.	Outbound	Cart status, abandonment events, coupon usage.
Analytics/BI (Tableau, Google Analytics 4)	Reporting on cart behavior, abandonment, and conversion.	Outbound	Cart events, abandonment/recovery flags, coupon redemption counts.

19. Reporting Requirements

- Daily cart abandonment rate dashboard, segmented by region (India, UAE, Southeast Asia) and channel (web, mobile app).
- Coupon/promotion usage and stacking-violation report for the marketing team.
- Stock-conflict frequency report identifying SKUs most frequently flagged as out-of-stock at cart or checkout stage.
- Abandoned-cart reminder performance report (send rate, open/click rate, recovery conversion rate).
- Average items per cart and average cart value trend report, by region and currency.
- Guest-to-registered cart merge conflict report for QA and product review.

20. Success Metrics

Metric	Current Baseline	Target (post-launch)
Cart abandonment rate	68%	≤ 55% within 2 quarters

Abandoned-cart recovery rate (via reminders)	4%	≥ 12%
Average items per cart	2.3	≥ 2.8
Cart-to-checkout stock conflict rate	7.5% of sessions	≤ 2% of sessions
Cross-device cart sync success rate	Not measured	≥ 99.5%
Cart page/API 95th percentile response time	3.4 seconds	< 2 seconds

21. Future Enhancements

- AI-driven personalized cart recommendations (e.g., "frequently bought together") surfaced within the cart view.
- Loyalty points redemption directly within the cart, once the loyalty engine build is completed.
- Predictive abandonment scoring to trigger proactive discount offers before a customer exits.
- Social/shared cart functionality allowing multiple customers to collaboratively build a single cart (e.g., gifting scenarios).
- In-cart delivery slot selection integrated with warehouse capacity signals from NetSuite WMS.

22. Glossary

Term	Definition
SKU	Stock Keeping Unit; a unique identifier for a specific product variant.
PDP	Product Detail Page; the page displaying full details of a single product.
PLP	Product Listing Page; a page displaying a list/grid of products, e.g. category or search results.
Cart Abandonment	When a customer adds items to a cart but exits without completing checkout.
ATP	Available to Promise; the quantity of a SKU confirmed available for sale after reservations.
GMV	Gross Merchandise Value; total value of goods sold through the platform over a given period.
B2C	Business-to-Consumer; a commerce model where a business sells directly to end consumers.
API	Application Programming Interface; a defined interface for systems to exchange data.
PII	Personally Identifiable Information; data that can identify a specific individual.
Coupon Stacking	The practice of applying more than one discount/coupon to a single cart or order.
Save for Later	A cart state allowing an item to be deferred from the active cart without deletion.

23. Appendix

Reference Documents:

- BRD-EC-CKO-002 — Checkout & Payment Module
- BRD-EC-PLP-001 — Product Listing & Discovery Module
- BRD-EC-PRC-004 — Pricing & Promotions Engine
- ShopSphere Platform Modernization Program Charter, FY2026

Revision History:

Version	Date	Author	Description of Change
0.1	10-Jan-2026	Vikram Singh	Initial draft covering scope and business requirements.
0.9	21-Jan-2026	Vikram Singh	Incorporated feedback from UX, QA, and Architecture reviews.
1.0	28-Jan-2026	Vikram Singh	Finalized for stakeholder review and sign-off.